CONSENT FORM FOR PARTICIPATION IN A RESEARCH STUDY

Study Title: Acquisition of Mathematical Concepts in the Human Brain

Principal Investigator: Dr. Marie Amalric

Researcher:

Faculty advisor: Pr. Samuel Gershman

Summary

This consent form describes a research study in which you are being asked to participate.

Why am I being invited to take part in this research study?

You are being invited to participate in this research study because its purpose is to compare how adults and typically developing children learn new mathematical concepts.

What should I know about a research study?

- Someone will explain this research study to you.
- Whether or not you take part is up to you.
- Your participation is completely voluntary.
- You can choose not to take part.
- You can agree to take part and later change your mind.
- Your decision will not be held against you.
- Your refusal to participate will not result in any consequences or any loss of benefits that you are otherwise entitled to receive.
- You can ask all the questions you want before you decide.

What is the purpose of this study?

The purpose of this study is to better understand how the human brain learns mathematics in school, and in particular how the brain changes as a result of learning. The broader goal of this research study is to advance the scientific understanding of conceptual development in humans through behavioral or neuroimaging activities.

How long will the research last and what will I need to do?

Behavioral activities will take 30 to 90 minutes including breaks. You can be invited to watch video clips and perform various tasks in which you will be asked to look at images, compare numbers, solve simple equations, read, or listen to sentences, draw geometric figures, answer reasoning questions.

Detailed information

What can I expect if I take part in this research?

Questionnaires. Prior to your first visit, as required by our funding source, we will provide you with a demographic questionnaire. The form does not include any personal identifying information and is given to you only so that we can assess the representativeness of the population in our studies. Your contribution to this

questionnaire is voluntary; if you choose to provide the requested information, your responses will be confidential.

Behavioral activities. Behavioral activities will take between 30 and 90 minutes including breaks and will involve various tasks in which you will be presented with a variety of math content. These tasks will be presented to you on a computer, on concrete materials such as paper or flashcards, and/or presented verbally by the experimenter. You will be asked to respond to questions verbally or by writing, by clicking buttons on the computer screen or using response buttons, or by drawing diagrams.

The session will take place in a private testing room in William James Hall.

Is there any way being in this study could be bad for me?

Behavioral activities. The risks and discomfort associated with participation in this study are no greater than those ordinarily encountered in daily life. You may request a break at any time. Your participation in this study is completely voluntary and you may refuse to participate or withdraw from participation at any time without penalty or loss of benefit to which you may otherwise be entitled.

Will being in this study help me in any way?

We cannot promise any benefits to you due to your participation in the study. However, we hope that you will enjoy this study, and your participation may improve our understanding of certain phenomena in the field of Developmental Psychology, as this study attempts to provide deeper insight into how the human brain learns about mathematics in school.

What happens if I say yes, but I change my mind later?
You can leave the research at any time it will not be held against you. Your participation in this study is completely voluntary and you may refuse to participate, or you may choose to withdraw from participation at any time without penalty or loss of benefit to which you are otherwise entitled. You will receive payment for the completed session even if you withdraw early. You may withdraw by informing the experimenter that you no longer wish to participate (no questions will be asked). The data already acquired from completed tasks may still be used for analysis in this project.

If I take part in this research, how will my privacy be protected? What happens to the information you collect?

You will be observed and recorded for data collection purposes only. The recording of your study session will be kept confidential and stored on a local server at our research facility, only accessible to lab personnel. All information coded from it will be associated with a code number rather than a name for us to identify it. Only researchers associated with this project will have access to the data.

In order to make research transparent to the scientific and broader community our study design and deidentified data may be made publicly available on open repositories such as the Open Science Framework (OSF). Deidentified study data means that all personal information about you (such as name, address, birthdate, and phone number) is removed and replaced with a code number. Sharing your deidentified study data helps researchers learn new and important things about brain functions more quickly than before.

Efforts will be made to limit the use and disclosure of your Personal Information, including research study records, to people who have a need to review this information. We cannot promise complete secrecy. Organizations that may inspect and copy your information include the IRB and other representatives of this organization.

Once all identifiers are removed from your identifiable private information collected during this research, that information could be used for future research studies or distributed to another investigator for future research studies without your additional informed consent.

What else do I need to know?

This research is funded by the European Research Council.

There will be no cost to you if you participate in this study.

You will receive course credits or $10/hour for taking part in behavioral activities.

Your participation is voluntary. You are free to stop your participation at any point. Refusal to participate or withdrawal of your consent or discontinued participation in the study will not result in any penalty or loss of benefits or rights to which you might otherwise be entitled. The Principal Investigator may at his/her discretion remove you from the study for any of a number of reasons. In such an event, you will not suffer any penalty or loss of benefits or rights which you might otherwise be entitled.

Who can I talk to?

If you have questions, concerns, or complaints, or think the research has hurt you, please don't hesitate to contact the lab for developmental studies at: 617-384-7777. You may also contact the principal investigator supervising this work: Marie Amalric, William James Hall, 33 Kirkland Street, Cambridge, MA, 02138, phone: 617-384-7777, email: mamalric@fas.harvard.edu

This research has been reviewed and approved by the Harvard University Area Institutional Review Board ("IRB"). You may talk to them at (617) 496-2847 or cuhs@harvard.edu if:

- Your questions, concerns, or complaints are not being answered by the research team.
- You cannot reach the research team.
- You want to talk to someone besides the research team.
- You have questions about your rights as a research subject.
- You want to get information or provide input about this research.

CONSENT STATEMENT:

Signature of the researcher obtaining consent: _____
Date: ___/___/___
Study ID: _____

The nature and purpose of this research have been satisfactorily explained to me and I agree to participate in the study described. I understand that I am free to discontinue participation at any time without any penalty, and that the investigator will answer any questions that arise during the course of research.

Participant's name (please print) _____

Participant's signature _____ Date: ___/___/___

Address to which we can send a summary in a few months: